

Project Definition Report

Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

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Summary

This Project Definition Report (PDR) contains a thorough description of the problem to be solved, including the context necessary to fathom the importance of solving it. It also outlines an initial miniature literature review that provides an overview of the most relevant work and sources to be further studied and cited. Finally, it covers the associated plan and timeline that will guide the project's execution, including a Gantt chart and key milestones.

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CHAPTER 1

Project Description

1.1 Background

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, currently consisting of only around 1,500 documents - significantly smaller than that of its younger counterpart, which now spans more than 6,000 documents.¹ But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.² Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael) (see Godart and Olivier (1976–1985)), these are only available in print form.

This motivated the development of the palæographical³ database *The Signs of Linear A* (SigLA), which seeks to "[develop] a systematic, exhaustive and user-friendly open access database of all Linear A inscriptions [...] in order to carry out statistical and palæographic analyses within the epigraphic corpus" (Salgarella and Castellan 2020). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known documents.^{4,5}

¹Steele 2023.

²Consiglio Nazionale delle Ricerche 2025.

³The term "palæographical" refers to the study of ancient writing systems.

⁴SigLA 2021a.

⁵SigLA 2021b.

1.1.1 A problem of manual digitization

The drawings currently stored on SigLA were made by Ester Salgarella using the painting tool Krita. For each document, she manually created her own annotated drawings based on the originals from Cefael. From these, she proceeded to create layered Krita files with metadata for each sign, which could then at last be imported to SigLA.⁶ According to her (personal communication), this has been a highly time-consuming undertaking. And the necessity of this kind of labor-intensive work has arguably been the biggest reason to why SigLA still does not contain the entire available corpus.

The aim of this project is therefore to investigate whether parts of this process can be automated. More specifically, whether some kind of semi-automatic segmentation tool can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database.

1.1.2 A problem of small data

The small size of the Linear A corpus and the irregularities in quality of its documents is a huge problem for the application of deep learning approaches, which typically require large amounts of data to perform well. Consequently, this project will instead seek to make use of generalized machine learning and computer vision, in particular Meta's state-of-the-art interactive segmentation tool, the Segment Anything Model (SAM) (Kirillov et al. 2023), which is pre-trained on a massive and diverse dataset of over 1 billion masks. SAM's generalization and interactivity (allowing user-placed markers that guide the model) might thus allow for semi-automatic annotation of Linear A inscriptions without the need for large amounts of training data, which is unfortunately not available in this context.⁷

⁶Salgarella and Castellan 2020.

⁷By the "interactivity" of SAM, what is meant is NOT any built-in user interface in the form of a GUI, but rather the possibility of prompt-based placement of rectangles of focus, and points of additions and deletions that guide the model's prediction.

1.2 Prior work / Literature review

This thesis places itself in the intersection of computer vision, digital humanities, and epigraphy (the study of inscriptions). Since it is a thesis of engineering, computer vision will remain the primary focus, both in terms of the research questions, and the methods used to answer them. As a result, the other two fields will merely provide the context and motivation for the research. The prior work and initial literature review in focus will thus concentrate on: 1. the relevant theory and state-of-the-art within computer vision (SAM, etc.), 2. the palæographical and digital humanities work that this project will seek to support (SigLA), and 3. the overall context thereof (Linear A).

1.2.1 Research context

For the context of Linear A and its corpus, the background section above can be referred to, citing the sources for: its size (Steele 2023), its new addition (Consiglio Nazionale delle Ricerche 2025), and where it's accessed (Godart and Olivier 1976–1985). And for the epigraphical and digital humanities work, the background section can also be referred to, citing the sources for SigLA: its paper (Salgarella and Castellan 2020), its about page (SigLA 2021a), and its document repository (SigLA 2021b). Here, additional context can also be provided by Ester Salgarella herself, as she is a co-supervisor of this project, and the main person behind SigLA.

1.2.2 Computer vision literature

As for the relevant theory and state-of-the-art within computer vision, the literature there can be further split into three parts; 1. sources of general image processing, 2. sources concerning SAM and its implementation, and 3. related computer vision script segmentation work for comparison.

For image processing, a general theory book (Gonzalez and Woods 2018) can be cited, as well as more method-specific papers (when other methods are used) like Otsu thresholding (Otsu 1979), and of course the OpenCV Python documentation (OpenCV Team 2024) for short implementation reference. Then, for SAM are: its paper (Kirillov et al. 2023), its lighter implementation MobileSAM (C. Zhang et al. 2023) - and the newest version of that one MobileSAMv2 (Zhao et al. 2023). Lastly, for related segmentation work there exists a paper using deep learning (in contrast to the methodology of this thesis) obtaining results better than state-of-the-art (P. Zhang, Li, and Sun 2024), and a lighter EU-funded historical document segmentation tool *dhSegment* (Oliveira, Seguin, and Kaplan 2018) also using deep learning.

1.3 Problem statement

As previously stated, the aim of this project is to investigate whether parts of the process of creating annotated drawings for SigLA can be automated. More specifically, whether some kind of semi-automatic segmentation tool using generalized computer vision - specifically the Segment Anything Model (SAM) - can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database. This is a problem that can be divided into the following three sub-tasks to be solved: 1. the pre-processing of original document images to make them suitable for SAM, 2. the incorporation of SAM into a segmentation tool potentially with the possibility of optional user interaction whose exports can be integrated into SigLA's workflow, and 3. the quantitative evaluation of the tool's performance compared to baselines against metrics of segmentation quality.

1.4 Empirical considerations

For the evaluation of the solution, a dataset will be compiled together with Ester Salgarella of at least 25 images of original documents from the corpus, selected and cropped from Cefael. These images are subject to Intellectual Property Rights (IPR) issues, and can thus not be displayed publicly if not through Cefael. However, they can still be used for the development and evaluation of the tool (and can be bought if necessary for show-casing within the thesis).

As ground truth, their corresponding digitized layered drawings, stored as Krita (.kra) files on SigLA (provided by Ester Salgarella) will be used and aligned for comparison with tool outputs. The ground truth data might then have to be cleared of any marked "deleted signs" or other annotation elements that our model is not intended to reproduce.

The solution will then be built as a pipeline, consisting of pre-processing, segmentation, post-processing, and export (e.g. as Krita files). At last, the primary deliverable of this project will be delivered to the external partner and co-supervisor Ester Salgarella in the form of a proof-of-concept semi-automatic annotation tool. All supporting components, including evaluation results, and analysis will be documented and presented in the written thesis.

1.5 Impact – innovation and application

The work of this thesis will present itself as a first proof-of-concept for the use of modern computer vision to accelerate the expansion of the SigLA database, supporting the further digitization of the Linear A corpus. Its main contribution will be a semi-automatic annotation tool, which might prove useful to epigraphers for generating segmentations of original document images, as they can then be completed and refined more easily. Hopefully, it will inspire further research and development

within the field of digital epigraphy, particularly in the application of generalized and interactive computer vision approaches such as SAM on Linear A, and perhaps even in the broader context of use on small corpus ancient writing systems where the lack of big data is ill fit for approaches using deep learning.

CHAPTER 2

Plan

2.1 Gantt Chart

Figure 2.1 shows an outline of the project plan, which extends 18 weeks (Feb 2nd - June 7th, 2026), including the studying, creation, and documentation of the pipeline in question.

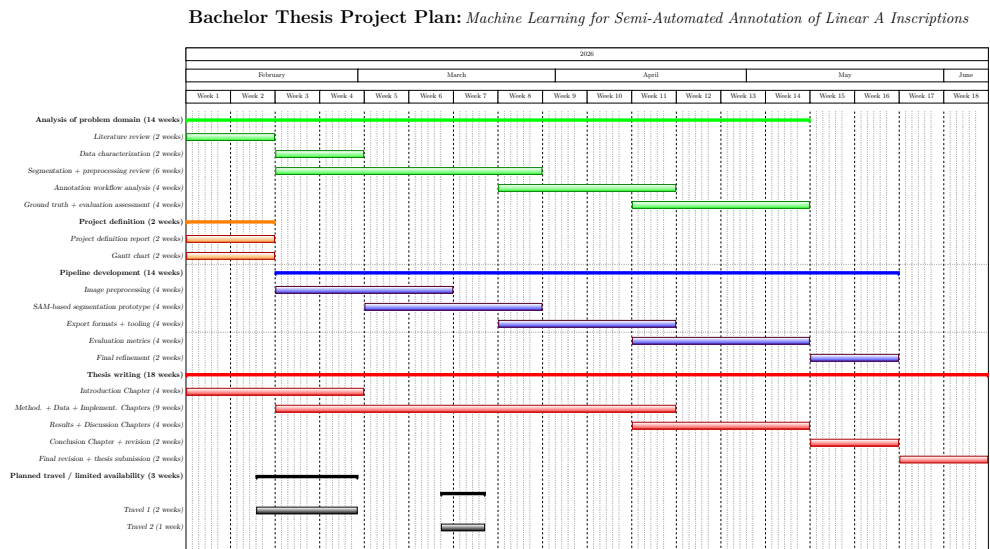


Figure 2.1: Project timeline (Gantt chart). See the appendix for a larger version (Fig. B.1).

This chart is divided into five groups of work packages, all complementing each other. These are (from top to bottom):

- in green, *Analysis of problem domain*: the study of the relevant literature and analysis required to proceed with the pipeline development. Work packages belonging to this group are placed in parallel with development, as they are necessary to inform it.

- in orange, *Project definition*: the writing up of this Project Definition Report (PDR) including its project plan and Gantt chart.
- in blue, *Pipeline development*: the creation of the segmentation tool pipeline to be delivered. In working packages, this includes the development of its three constituent modules (preprocessing, segmentation, and export + tooling), the definition and implementation of its evaluation metrics, and its final refinement.
- in red, *Thesis writing*: the simultaneous writing up of the thesis, including its chapters, revision and submission. The method, data and implementation chapters will be written together in parallel with pipeline development, while the results and discussion chapters will be mostly written during work on evaluating the pipeline.
- in black, *Planned travel / limited availability*: the periods where the author will have limited availability due to travel. These are included, as they might affect the project timeline, and have been taken into account in the planning of the work packages.

The in-group work packages partly overlapping in time, signals that they might have to be carried out in parallel during the transitions between them.

2.2 Milestones

Although each work package functions as its own milestone, deliverable and deadline, there are a few key milestones in the project that are especially important to keep track of. These are summarized in Table 2.1.

Table 2.1: Milestones. D: Deadline TH: Thesis.

#	Title	Description	Type	Week
M_1	Project Definition Report	Approval of the PDR and Gantt Chart.	D	4
M_2	First full prototype	Creation of the First Pipeline Prototype.	D	11
M_3	Final pipeline	Creation of the Final Pipeline.	D	16
M_4	Thesis written	Finalization of Thesis content.	TH	16
F	Thesis hand-in	Hand-in of final Thesis.	TH	18

Here, the milestones M_2 and M_4 act as buffers to ensure that the project is progressing as planned, and that the final hand-in deadline can be met. Thus, there is made space for possible delays in the project timeline, should any unforeseen issues arise during the development of the pipeline or writing of the thesis.

APPENDIX **A**

Acronyms

SAM Segment Anything Model

Cefael Collections de l'École française d'Athènes en ligne

PDR Project Definition Report

IPR Intellectual Property Rights

APPENDIX B

An Appendix

Bachelor Thesis Project Plan: *Machine Learning for Semi-Automated Annotation of Linear A Inscriptions*

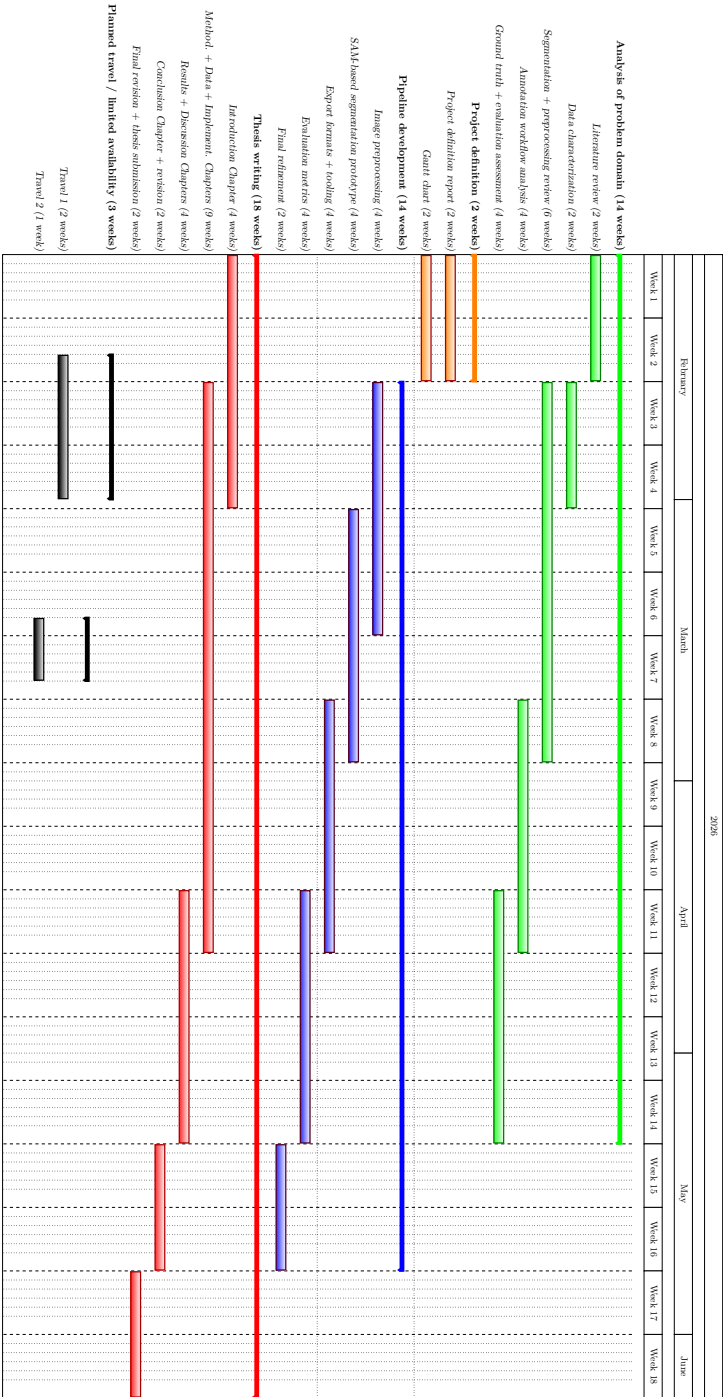


Figure B.1: Extended project timeline (Gantt chart). Larger version of Fig. 2.1.

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